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Internet Of Things Interface Using Augmented Reality: An Interaction Paradigm using Augmented Reality

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Abstract:The IOT (Internet of Things) covers all devices connected to the network that use the Internet of Things and the data sharing over the network. IOT is now turning into today's rapidly expanding technology and the future is glowing. Home Automation Systems is the most popular application of IOT. The recent training in IOT has now become IOE (Internet of everything). With this technology, one can use any application to access each machine, equipment and machinery in its vicinity. Each home will in future become an intelligent home, where users have remote access to all the appliances in their homes. The Augmented Reality (AR) is one of today's top trend technologies, namely the creation and visualization of digital solutions to problems in the real world. In the physical environment AR lets us view digital data in order to bridge the gap between the physical world and the digital world. AR utilizes the camera of any smart phone and the application utilizes the phone camera to show the digital environmental data before the consumer. There have been substantial study works on IOT. Security has been worked on by some researchers to improve data sharing encryption. Some tested several IOT-generating wireless systems. But the paradigm of communication is not concentrated. With regard to IOE, millions of IOT devices are going to be available and think about interacting with them with the ancient middleware application paradigm. The user cannot remember which specific interface which device is controlling. Moreover, the whole scope of IOT remains uncertain, although IOT is accessible and used, but yet additional characteristics have not been fully digitized. The purpose of our research is to merge Augmented Reality's strong digitization with IOT. Our study is also to add another digitized IOT function, which is digital wires rather than physical cables. With this strategy we can connect inputs to one or more outputs using digital IOT cables via AR, just by linking digital I / O points that we developed in our project. The physical function of any device may also be shared with another physical device, such as sharing the function of an oven timer switch with any device, such as a fan, or using a fan controller, not only to manage the fan velocity but also to digitally link it to a light bulb and regulate the intensity / brightness of light via regulator.

Keywords: Internet Of Things; Augmented Reality; Home Automation; Mobile Application;

I. INTRODUCTION

Internet In order to share information and be part of the network, Internet Of Things (IOT) has been introduced by various wired or wireless connections. These are GSM, Bluetooth, Wi-Fi, Ethernet and Zigbee techniques. Many researchers have been conducted in the IOT sector, some have studied connection methods, others IOT safety, some infrastructure. Another advance technique Augmented Reality (AR) has been launching a fresh dimension of computer vision (CV), which is now becoming a fresh study subject. India recently proclaimed a plan to redesign its metropolitan areas as "100 Smart Cities" and China has launched its activity "Smart Cities, Smart Growth" [1]. More than 54% of the global population resides in metropolitan areas, which is expected to decrease to 66% by 2050. There is a large number of people moving continuously to urban regions and the pace and magnitude of urbanization creates fresh hosting, transport and vitality problems. Treating urban communities is a cross-national experiment because urban consume 70% of the vitality they create and supply up to 80% of global ozone depleting substances [2]. The description of the system in [3] is based on the GSM Module that utilizes the radio signal to transmit and collect information and controls the linked equipment from a microcontroller. In a tiny region of the space, intelligent home automation as outlined in [4] can be used regionally. Using this system users must attach his smart phone to the Bluetooth device he wishes to manage, the smart phone application operates as the system middleware. This system's safety is based on the Bluetooth model, but the safety is much smaller for the lowest version and can readily be accessed because the information is not encrypted during sending and receipt. Implement IOT with the wireless private network and Android application to access all systems. The system presented in [5] allows users, through a single WLAN router scheme, to interact or regulate up to 10 devices. It is also relatively small and cheap. There are two definitions in the research [6]. The first definition of "Augmented reality" is artificial, improved environment. Characteristics of the augmented reality system are provided and innovations are classified in this system. Secondly, educational potential exists in this context. The study [7] reflects a head up display of AR. For example, the projection of optical routes via simulated components on a virtual optical bench and the implementation in which a remote user controls a laser point that another user uses to imply the items concerned.

Two distinct directions were taken for this literature study. First presents the methods of IOT implementation and second presents the significance, application and future of AR. The literature review showed that IOTs like Bluetooth, GSM, WLAN and Over the Internet were implemented in many respects. After evaluating existing IOT schemes, the AR application together with IOT were provided as an interface to enhance the communication and to change the framework of the IOT so that the IoE of the future can readily reach it. AR is known for its more advanced and readily available depiction of digital data, such as interfaces. AR offers a way to readily identify items and users do not have to save them. The same method must be used with IOT as a user must prevent memorizing and communicate with objects immediately, as well as communicate with objects of the physical globe. AR also provides a solution to improve and expand IOT data digitalization.

II. Related Work

This article examines some associated work to highlight implementation of the augmented reality.

White et al. [8] explained augmented reality uses in context of IoT to provide information to users and providers and give a description of an AR application. Kasahara et al. [9] proposed a system in which users are free and able to tag, create, , and share data around via a remote server. Iglesias et al [10] proposed a system that is about context-aware representation for objects we used in our daily life with respect to the proximity of user, user's attribute, visibility and orientation. Ullah et al [11] proposed a touch system for smart home. In which the user touch the object seen on a display. Marques and Pitarma [12] describe a system for real-time monitoring of indoor air quality. The system consists of the ESP 8266 as a communication and processing unit and the MICS-6814 sensor detection unit. The System also provides smart phone applications for data consulting and real-time notification.

III. Tools & Technologies

A. Software Tools

1) Unity 3D

Unity is a multi-platform gaming engine created for computer games, consoles and cellular telephones by Unity Technologies [13]. First proclaimed for OS X only, it has since been extended to target 27 Platform at Apple's Worldwide Developers Conference, held in 2005.

2) Vuforia SDK

Vuforia [14] is the SDK (Software Development Kit) for cell phones which supports the development of apps for increased reality. It uses innovation from Computer Vision for the gradual perception and follow-up of flat images, such as boxes and simple 3D objects. This capability allows technicians to position and place virtual papers, for instance 3D models and other media, with real images when viewed through a cell phone camera. The virtual issue then increasingly monitors the position and introduction of the image, thus comparing the point of perspective of the protest to the point of perspective of the image target and giving rise to the concept that the virtual issue is a part of that scene of actual practice.

3) PubNub SDK

Data Stream Network PubNub is the secret sauce that makes everything function [15]. PubNub DSN connects 15 redundant presence points worldwide to a single network that can handle hundreds of millions of concurrent devices and relay trillions of emails. Think of PubNub, however, as the web-based content delivery network (CDN). A Data Stream Network (DSN) is optimized to serve dynamic content (that is' data in movement') in real time while a CDN is intended to serve static content. PubNub can be used to move tiny emails to one or multiple systems rapidly (in the main smart phones, browsers, microcontrollers, tablets, etc.) just about every device that can connect to the web TCP / IP as well as back up for two-way interconnections.

B. Hardware Tools

1) Arduino Micro Controller

Arduino [16] is a platform for the development of electronics applications in open source. Arduino comprises of both an integrated development environment (IDE) that operates on your desktop, and a physically compatible circuit board that is often referred to as a microcontroller.

2) ESP8266 Wi-Fi Module

The ESP8266 [17] is a 2.4 GHz Wi-Fi system (802.11 b / g / n, WPA / WPA2 supports), general purpose input / output (16GPIO), I2S (GPIO sharing pins), UART (on specialized pins and on GPIO2 a transmit UART is allowed). ESP8266 is a computer system with capacities for 2.4 GHz Wi-Fi (802.11 b / g, WPA 2 supports). Figure 1 indicates the ESP8266 WiFi pin setup.

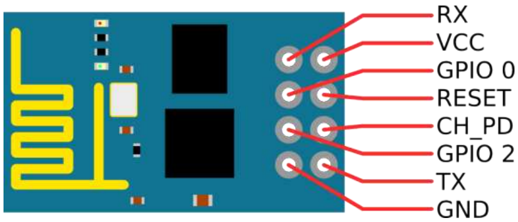

Figure 1. ESP8266 Pin Configuration

3) DC Devices

The electrical circuit DC solution is the solution that contains all the voltages and current. The DC component can be described as the predicted value, or the average value of the voltage or current over all times, as the total value or the value of the voltage or current can be decomposed into a sum of DC components and a zero mean time component. Uses of DC devices are preferred for varying value of any input device without use of any voltage regulation system. A DC Bulb, Fan or any other device will always work on same range of current value.

IV. Implementation

A. System Design

1) Hardware Design

Hardware Design is composed of two main components. One is to create a micro controller's IOT device, and another is to regulate the DC machine with a micro-controller enabled by the IOT. The schematics of the circuit design are shown below.

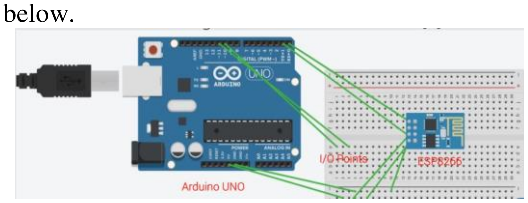

Figure 2. Enabling IOT using Arduino Micro-Controller

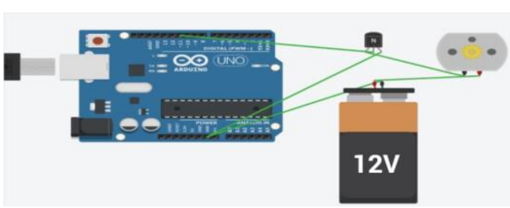

Figure 3. DC Device Controlling via Arduino

a) Arduino Flowchart

The process of Arduino is split into two components. Input devices are one control, and in output devices are other controlling. The following flowcharts show both processes.

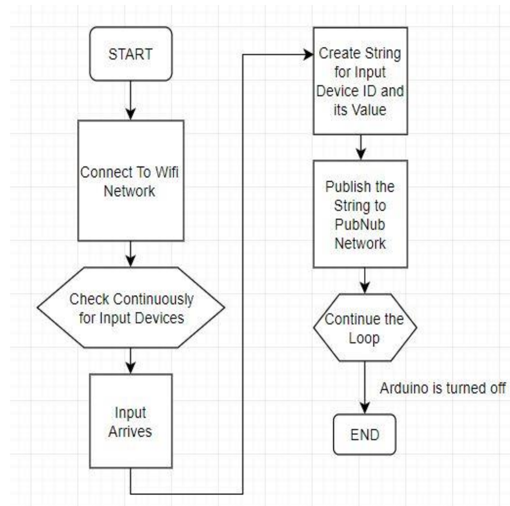

Figure 4. Input points flowchart

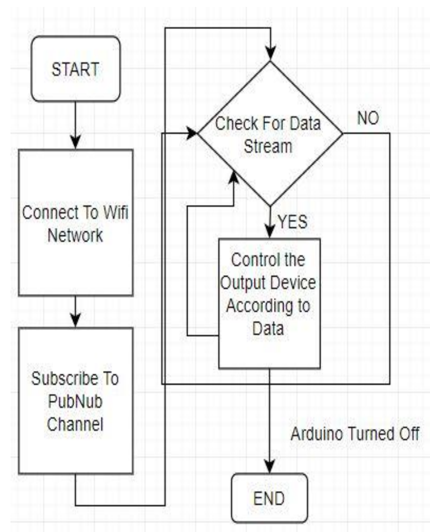

Figure 5. Output points flowchart

2) Software Design

The Mobile Application is developed using the Unity 3D Control Tool for Cross-Platform Development. The Design section comprises the design of trackers and the mapped objects using SDK Augmenting Realities shown below.

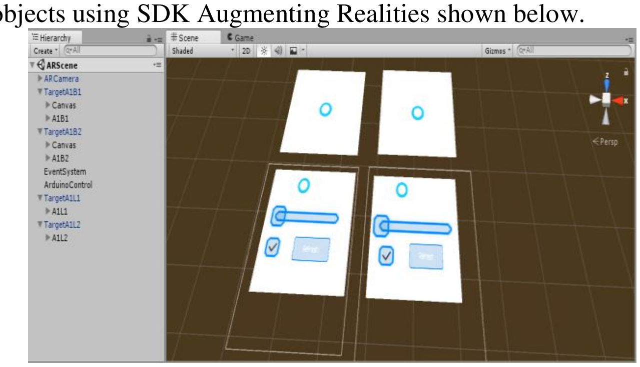

Figure 6. Objects Placement in Unity 3D environment

a) Unity 3D Program Flowchart

Using this platform, mobile applications are created. The general application workflow is shown in the following flowchart.

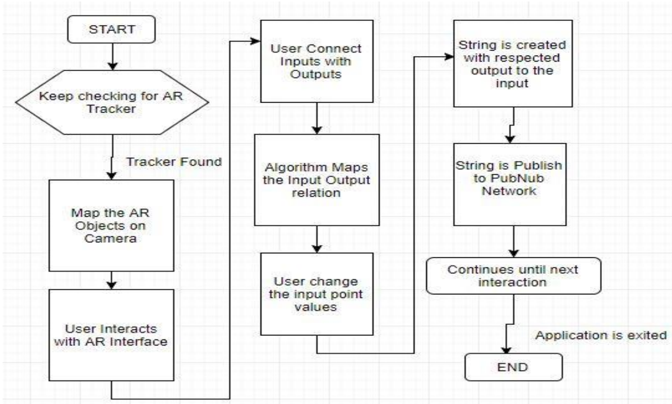

Figure 7. Mobile App Flowchart

3) Complete System Process Design

The entire system is split into 3 components, 1 is hardware, 2 is software and 3 is networks. The PubNub api is used to make the networking portion. The following flowchart shows the method.

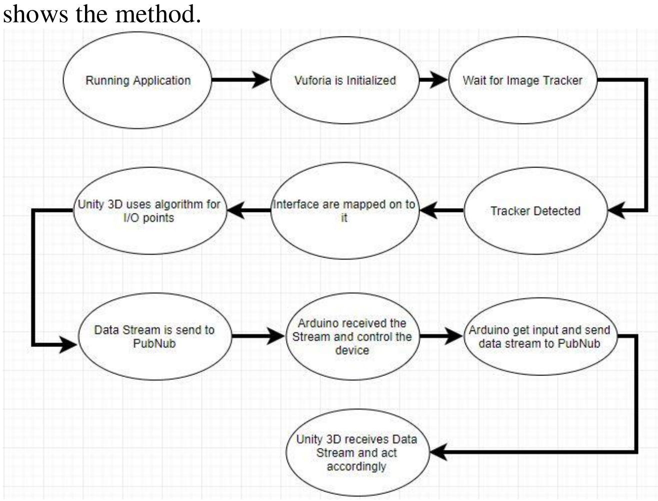

Figure 8. Objects Placement in Unity 3D environment

B. Proposed Algorithms

The idea of the digital cables is our work for this project. The digital cables are based on the input and output AR I / O point objects and line renderer. We communicate with just two kinds of electronic appliances, either input or output, on a daily basis. This idea now means that we have broken IOT devices into two "input point device" and "output point device" classifications. Now that this interaction is taken into account, we have come to an important point: "One single input can control several outputs but only one single output can control one input." This declaration is the cornerstone of our algorithm. We have now intended the software algorithm to operate and alter the notion of digital cables according to this idea.

1) Algorithm for Digital Wires Illustration

Digital cables in mobile application are illustrated in two components. Two kinds of artifacts exist. Input points are one, and output points are another. The pseudo codes are described below for both objects.

a) Output Objects Pseudo Code

STEP 1: Start
STEP 2: Create a method which will be called for initialization
STEP 3: Initialize the Line renderer object which is attached with the Output object
STEP 4: Create a Global Variable to store the game object value.
STEP 5: Create a callback method which will be called when output object is clicked.
STEP 6: In the callback method we are checking that whether the Global Variable has stored any game object value or it is null
STEP 7: If global object has value then we have to move that value to a local variable and make the global variable null
STEP 8: Create a method which will be called every frame per second and update the unity script
STEP 9: In the updating method we check that whenever there is a value stored in the local variable declared previously we will re initialize the Line Renderer to start from the current object to that game object which we have got from global variable hence creating a line (wire) between these two objects
STEP 10: End

b) Input Objects Pseudo Code

STEP 1: Start
STEP 2: Create a callback method which will be called when input object is clicked.
STEP 3: Whenever mouse is clicked on the input object we just put that input object into the Global Variable hence to be used in the Output object script.
STEP 4: End

2) Algorithm for Working Implementation

For backend work of digital wires the algorithm is used to write the 3D unity script.

a) Global Object Pseudo Code

STEP 1: Start
STEP 2: Create Global Variable of Dictionary (A key value pair data type) in C# script
STEP 3: Create a global variable of String Array

STEP 4: Create a global accessible method with arguments of the input name which is calling and the value which the input needs to provide
STEP 5: Then we have to check whether how much outputs the specific input has been connected with using line renderer
STEP 6: Then we put all the outputs the input object is connected in the temporary string array
STEP 7: End

b) Output Object Pseudo Code

STEP 1: Start
STEP 2: Create a callback method to check for mouse input on output object
STEP 3: Whenever the output object is clicked we check if the output is previously connected with any other input object or not by check the Global Dictionary
STEP 4: If it is already there then we will remove it first from that connection
STEP 5: After that we will put the output object and the input object in the Dictionary as Key Value Pair
STEP 6: End

3) Algortihm for Network side Implementation

The data is transmitted through the PubNub network between a mobile software implementation and IOT hardware devices. The algorithm is used to communicate with each other on both the hardware and software side of the scheme.

a) PubNub Subscribe Pseudo Code

STEP 1: Start
STEP2: Create PubNub thread for continuous communication with the PubNub network and for receiving and sending Data
STEP 3: Using the PubNub thread for Global Object script to send the encoded string
STEP 4: Create a separate script for PubNub Subscribe in order to receive data
STEP 5: Subscribe to the particular channel on which all devices are sharing data
STEP 6: Check for the encoded string data received from the network
STEP7: Call the global object method for checking the related outputs
STEP 8: End

b) PubNub Publish Pseudo Code

STEP 1: Start
STEP 2: Create object of PubNub Thread Class
STEP 3: Provide the Generated keys of Publishing and Subscribing to thread object
STEP 4: Using PubNub thread subscribe to the particular channel
STEP 5: Publish data whenever it is required by calling in any script
STEP 6: End

4) Algorithm for Hardware Side Implementation

In order to connect to the network and communicate with other nodes in the network, we used Arduino Micro Controller and control systems physically connected to Arduino. Each Arduino input or output device is an IOT system that shares or receives information from the network via Arduino.

a) Input Points Pseudo Code

STEP 1: Start
STEP 2: Initialize Input object as digital input and assign Pin Number
STEP 3: In the Arduino loop method continuously check for high and low input
STEP 4: Whenever the value is high we write the Button ID and High value and when its low we write Low value line in order to serial line for Esp8266
STEP 5: For analogue inputs we first need to map it with outputs levels
STEP 6: After then we have to write that Input ID along with output value
STEP 7: End

b) Output Points Pseudo Code

STEP 1: Start
STEP 2: Initialize digital output or analogue output accordingly and assign pin number
STEP 3: In the Arduino Loop method we need to run a loop to check the receiving data stream from the network and iterate over the stream to access the characters
STEP 4: In the stream, we need to check for encoded string for Output ID and value
STEP 5: After checking we need to send that particular value to the particular output
STEP 6: Clear the current stream and wait for another one to receive from the network
STEP 7: End

c) Networking Pseudo Code

STEP 1: Start
STEP 2: Make a connection with network using the previous algorithm
STEP 3: Connect with PubNub using the Generated Keys for Publish and Subscribe
STEP 4: Check if data is available and get the data stream using PubNub object
STEP 5: Provide the Data Stream to output Algorithm if Node is Output, keep in mind that we can only create one Arduino either as input or either as output

STEP 6: For Input Node, use the PubNub object to publish the data of Input Points Algorithm
STEP 7: End

V. Conclusuion & Future Work

In the Internet of Things, we have used AR effectively as a fresh interaction paradigm. The AR can be used easily for any appliance, whether it is an instrument or an object. As the notion of digital hoses, our suggested study has been carried out. The electronic cables have not only been developed as part of the software but also as a programmed network, equipment and digital links. In this scheme, there are several things that can be enhanced. Few of these potential projects use Tracker less AR than AR based on the tracker. This enhancement will contribute to the removal of the tracker requirement. In the picture below, the final results of the hardware and software were shown.

Figure 9. Hardware Result

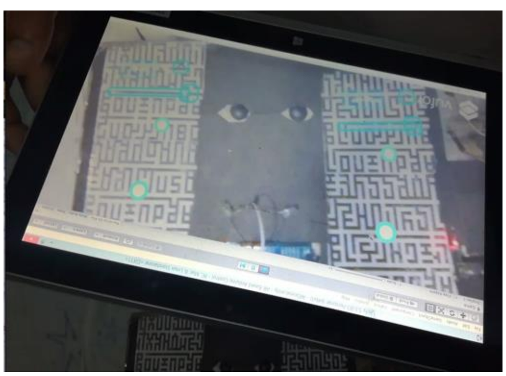

Figure 10. Software Result

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